

Liberty File Analyzer and Merger

User and Developer Documentation

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1 Introduction

This document provides complete documentation for the Liberty File Analyzer and Merger tool. The software allows engineers to:

- Parse Liberty `.lib` files into structured representations.
- Analyze cell, pin, timing, and power data.
- Merge multiple libraries or multiple cells into a consolidated output.
- Interact easily through a graphical user interface (GUI).

The tool is written in Python and is intended for VLSI/ASIC designers working with Liberty format files.

2 Installation

1. Clone the repository:

```
git clone https://www.github.com/xiayu87/mrglib
cd mrglib
```

2. Install dependencies:

```
pip install -r requirements.txt
```

3. Run the GUI application:

```
python gui_app.py
```

3 Graphical User Interface (GUI)

The GUI provides the primary way to interact with the tool. It was implemented in `gui_app.py`.

3.1 Main Features

Load Library Opens a file dialog to select one or more Liberty files.

Analyze Parses the selected file(s) and displays structured information on cells, pins, timing, and power.

Merge Combines multiple Liberty files (or multiple cells) into a unified library.

Export Saves the analyzed or merged result into a new file (`merged.lib`).

Logs/Output Shows status messages and error feedback.

3.2 Buttons and Parameters

Each button in the GUI has a dedicated function:

Button	Functionality
--------	---------------

Load File(s)	Opens dialog to choose Liberty files. Multiple selection allowed.
Analyze	Runs parser on loaded files, producing a detailed structural view.
Merge	Combines loaded files into a new internal representation.
Export	Writes out the merged structure into a valid <code>.lib</code> .
Exit	Closes the application gracefully.

4 Configuration Files

The tool allows customization through both `config.py` and `config.yaml`.

4.1 Important Parameters

postfix.regex Regular expression used to detect postfixes in cell names.

precedence Rules for resolving duplicate attributes (e.g., “first” or “later”).

units_policy Defines which units to adopt when conflicts exist (options: “first”, “later”).

rewrite_attr_equals_cellname Boolean option; if true, rewrites internal attributes to match the cell name.

4.2 Editing Config

Both YAML and Python formats are supported. The YAML file is preferred for non-developers.

5 Module Documentation

5.1 `parser.py`

- Reads Liberty `.lib` files.
- Builds structured dictionaries for cells, pins, timing, and power.
- Provides error checking for malformed syntax.

5.2 `merger.py`

- Combines multiple parsed Liberty files into one structure.
- Ensures no duplicate cell names.
- Resolves conflicts using precedence rules from configuration.

5.3 `analyze.py`

- Inspects parsed library objects.
- Produces summary tables of cells, pins, timing arcs, and power entries.

5.4 `serializer.py`

- Converts internal data structures back into valid Liberty syntax.
- Handles formatting, indentation, and attribute ordering.

5.5 `util.py`

- Contains helper functions for string manipulation, regex matching, and validation.

5.6 `config.py`

- Provides default configuration values (used if YAML not present).

5.7 `gui_app.py`

- Implements GUI window and event bindings.
- Integrates all functions: parse, analyze, merge, export.

6 Command-Line Interface (CLI)

Note: This section is under development.

The codebase contains placeholders for CLI commands such as `analyze` and `merge`, but they are not stable yet. For now, only the GUI workflow is officially supported.

7 Output Files

7.1 Analyzed Results

Displayed in GUI and optionally saved as structured JSON.

7.2 Merged Library

Produced as `merged.lib`, guaranteed to follow Liberty format.

8 Error Handling

Common issues:

- **Malformed input:** Parser will log a warning and skip invalid blocks.
- **Conflicting attributes:** Resolved using precedence rules.
- **Empty merge:** If input libraries share no compatible cells, merged output may be empty.

9 Tutorial

Following is a step by step tutorial to merge cells in a library.

9.1 Main Window

The GUI provides a visual interface for interacting with Liberty files. Users can load input files, analyze them, merge cells, and export results.

9.2 Loading Input Files

The user can select one or multiple Liberty files for parsing. The selected files are listed in the interface for further processing.

9.3 Analysis

The **Analyze** button triggers the analysis of selected files. This checks syntax validity, cell completeness, and reports errors.

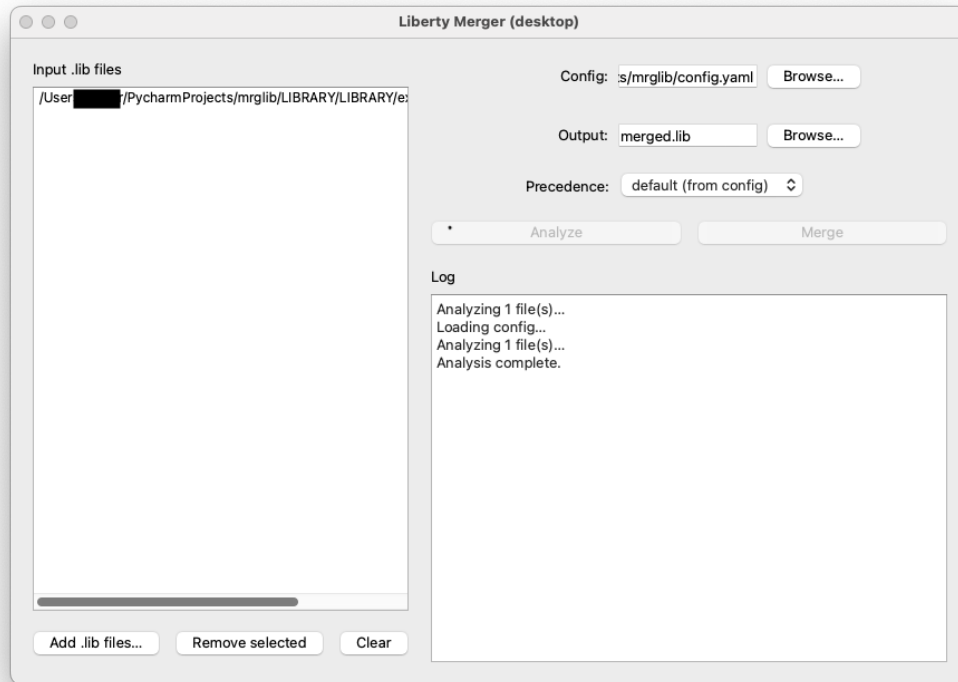


Figure 1: Main GUI window on launch.

9.4 Merging Cells

The **Merge** button allows merging multiple cells across files, or merging subcells within a single file. The merged result is displayed in a text area.

9.5 Exporting Results

The **Export** button saves the merged Liberty file into a specified directory.

9.6 Parameter Settings

Configuration options are available under the **Settings** menu. These include library type selection (CCS, ECSM), merge rules, and error handling preferences.

9.7 Progress Monitoring

The tool provides progress bars during heavy operations (e.g., parsing large libraries).

9.8 Results Summary

After merging, a summary panel displays statistics such as number of cells merged, timing arcs retained, and power models combined.

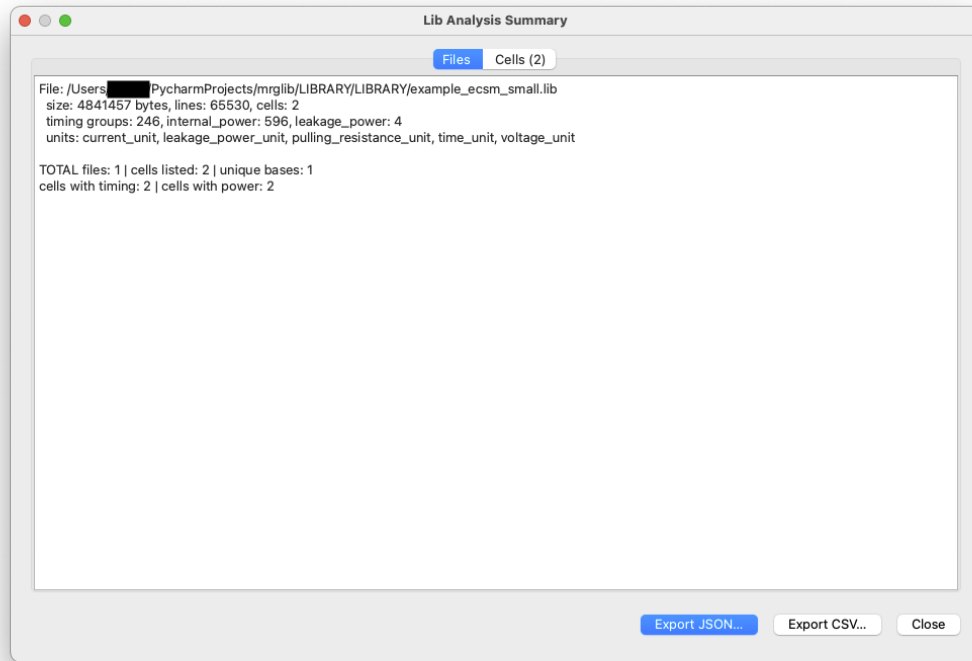


Figure 2: File selection dialog for Liberty files.

10 Future Work

- Complete CLI command set.
- Add visualization of timing/power graphs.
- Expand error reporting to pinpoint faulty cells.

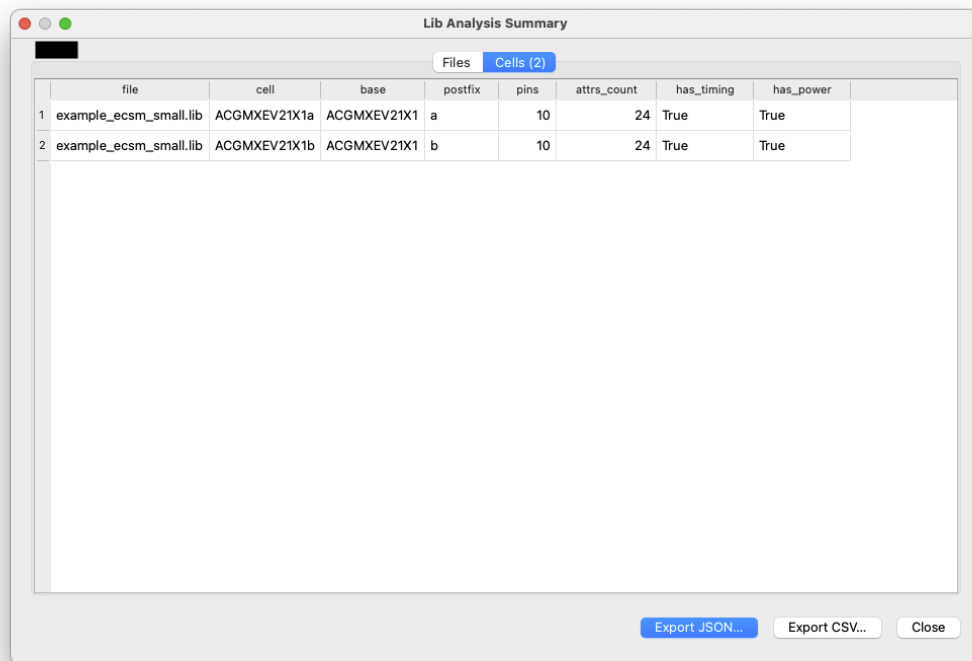


Figure 3: Analysis results displayed in GUI.

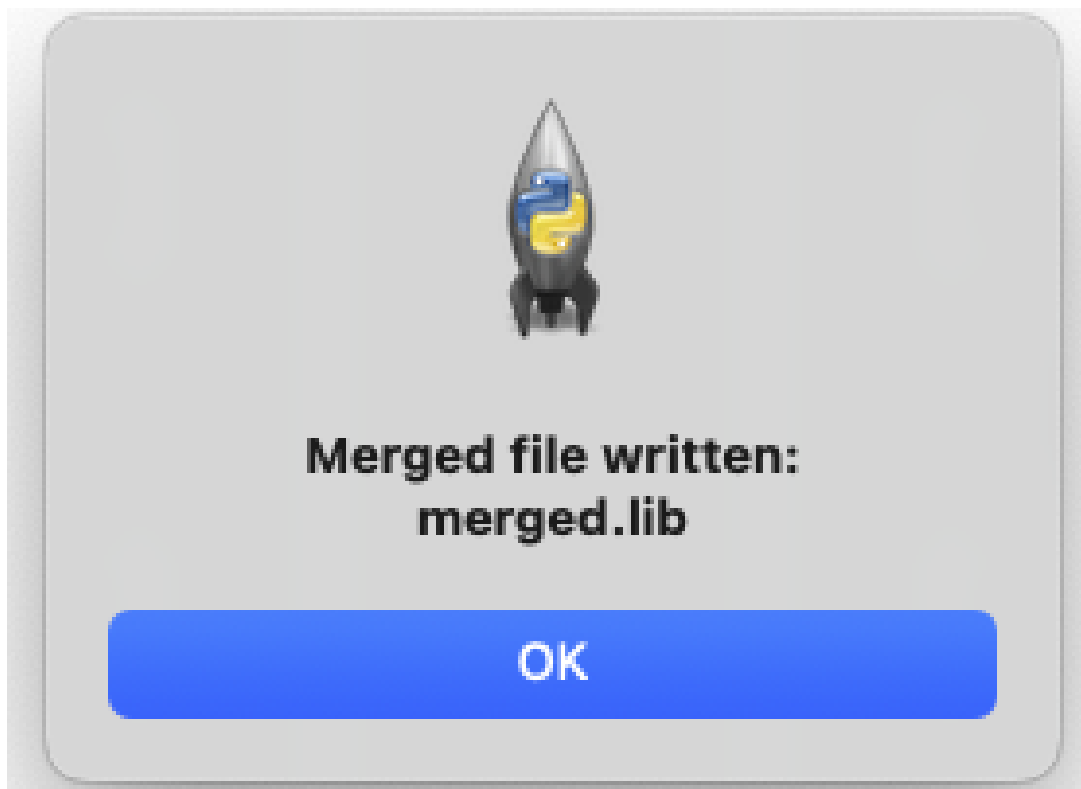


Figure 4: Merged cells preview in GUI.

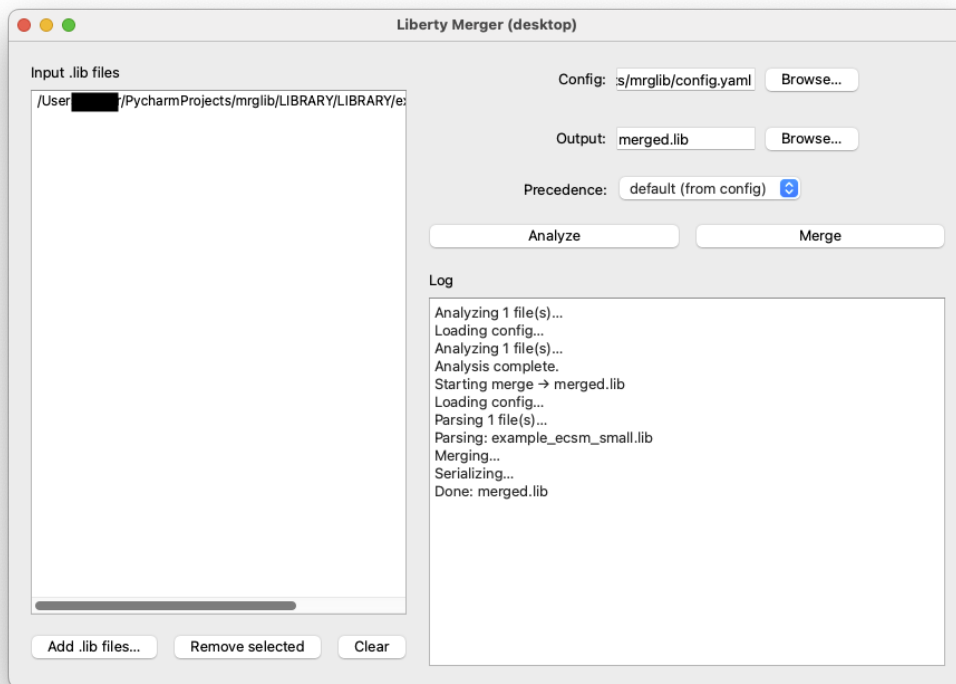


Figure 5: Export merged Liberty file.

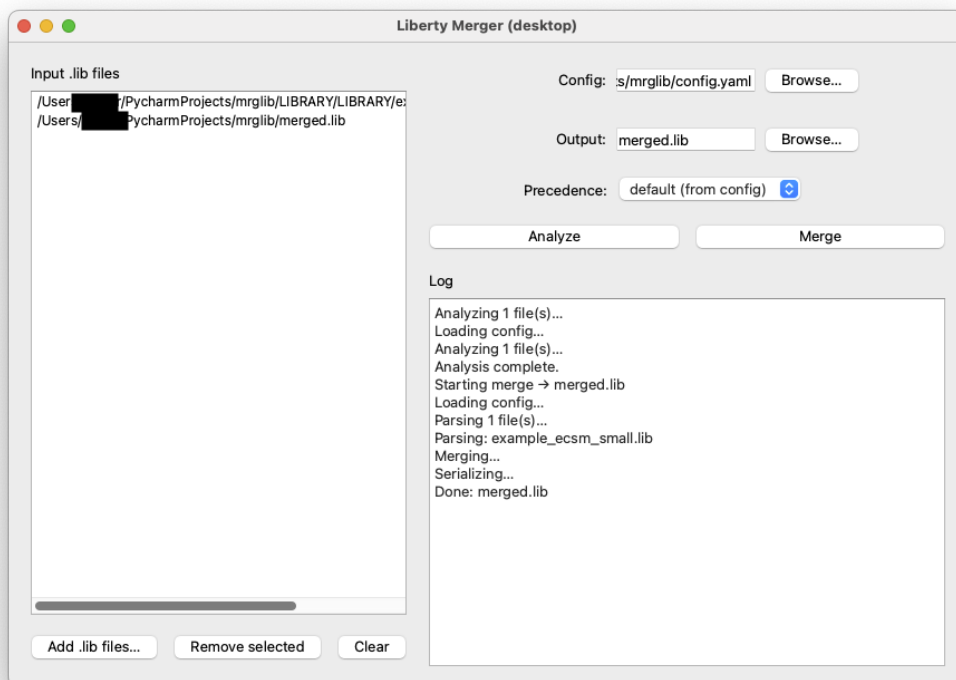


Figure 6: Configuration settings panel.

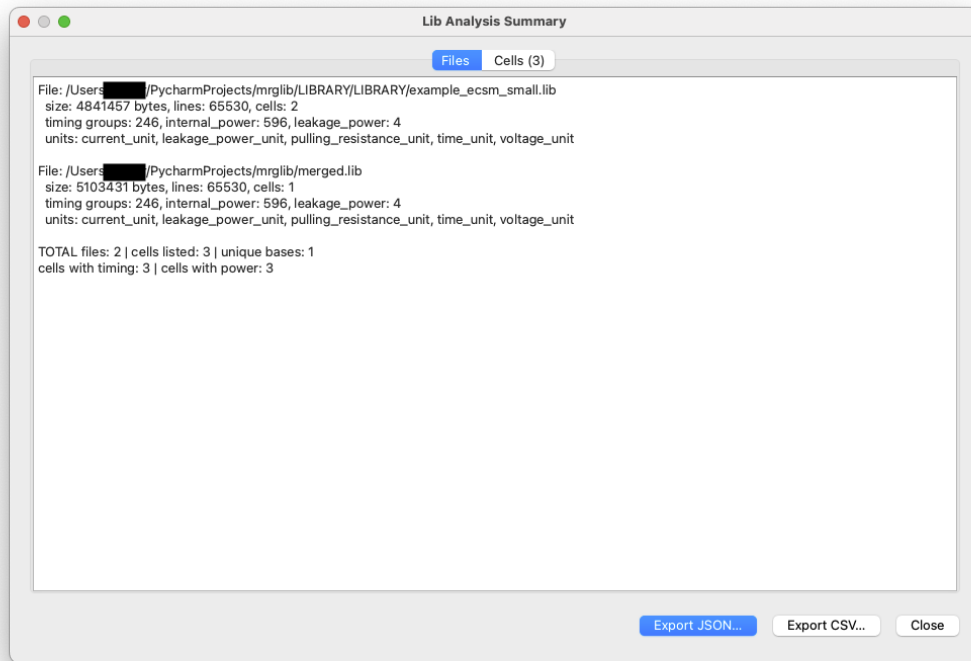


Figure 7: Progress monitoring during merging.

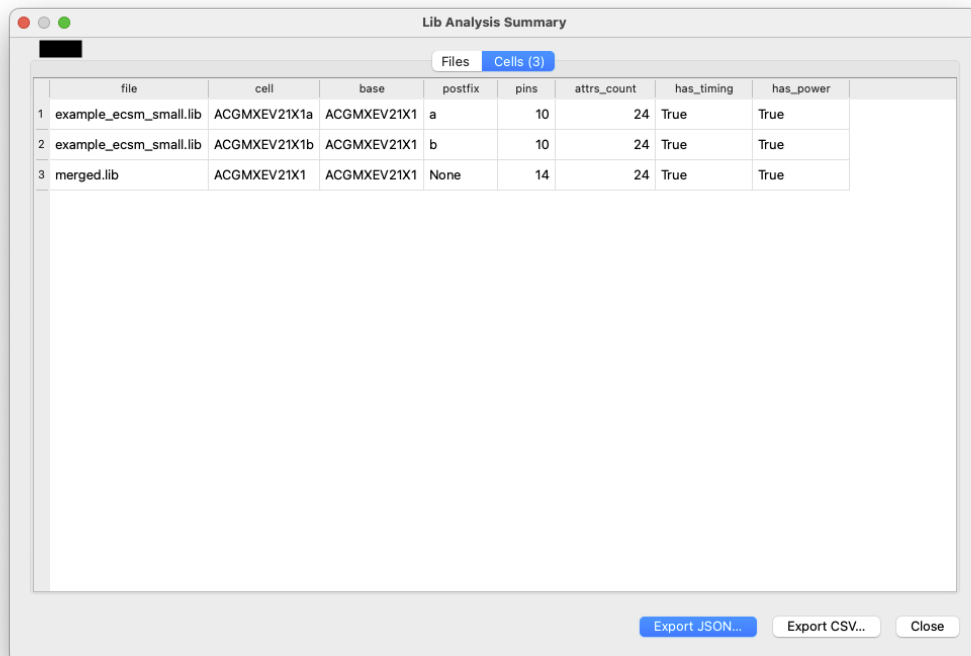


Figure 8: Summary of merged results.